

Information Asymmetry and the Propagation of Failure: A Script Analysis of Redundant Attempts in Shared Criminal Data Pools

Abstract

The migration of illicit activities to online spaces has facilitated the rise of shared criminal data pools, where offenders trade tactics, targets, and vulnerabilities. While these environments significantly lower the barrier to entry for prospective criminals, they are hindered by severe structural inefficiencies. This paper explores the phenomenon of "silent failures" within illicit networks. Integrating rational choice theory and crime script analysis, this paper posits that when an individual fails in a criminal attempt based on shared intelligence, structural disincentives prevent them from communicating this failure back to the data pool. Consequently, this persistent information asymmetry causes subsequent offenders to overestimate the probability of success, leading to predictable, redundant, and identically flawed criminal attempts.

Introduction

Criminal networks increasingly rely on the aggregation of shared knowledge. From dark web forums to encrypted peer-to-peer communications, offenders organize around shared data pools that distribute target profiles, exploit kits, and actionable intelligence (Kloess & van der Bruggen, 2021). These repositories act as digital hubs that distribute the "scripts" necessary to execute complex crimes.

However, unlike legitimate organizations that utilize feedback loops to optimize operational efficiency, illicit networks suffer from chronic information asymmetry (Sebagh et al., 2022). A critical gap in criminological literature is the analysis of how these networks process *failure*. When a criminal attempt fails, the offender rarely broadcasts this outcome. Without a corrective feedback mechanism, the original, flawed data remains active within the shared pool. This paper investigates how this absence of failure reporting influences the decision-making of other network participants, driving redundant attempts that inevitably end in identical failure.

Theoretical Framework

Rational Choice Theory and Expected Utility

The foundational premise of rational choice theory is that human behavior, including criminality, is driven by an assessment of competing alternatives under conditions of scarcity (Mathis & Steffen, 2015). Under this model, an offender weighs the anticipated rewards of a criminal act against its potential risks. We can formalize the expected utility of a given crime $EU(C)$ using the following expected value equation:

$$EU(C) = p(S) \cdot U(S) - p(F) \cdot c(F)$$

Where $p(S)$ represents the perceived probability of success, $U(S)$ represents the utility or payout of a successful attempt, $p(F)$ represents the perceived probability of failure, and $c(F)$ denotes the costs or consequences associated with failing.

Crucially, an offender's assessment of $p(S)$ and $p(F)$ is heavily reliant on the information available within their network. Rational choice theory often highlights that humans do not act with perfect rationality, but rather with "bounded rationality," relying on limited or skewed information (Mathis & Steffen, 2015).

Crime Script Analysis

Crime script analysis provides a structured methodology for understanding the sequential steps an offender must take to successfully execute an offense. These scripts break down an event into logical stages—such as preparation, entry, execution, and exit (Ho et al., 2024). In modern cybercrime and organized fraud, these scripts are commodified. Offenders package successful routines and sell or share them within dark web marketplaces and encrypted forums. However, if a script becomes obsolete or a target develops adequate defenses, the script's utility degrades.

The Ecology of Shared Data Pools

Trust and Obscurity in Illicit Networks

Shared criminal data pools operate under unique constraints. Participants must navigate an environment characterized by near-complete anonymity, where the true identities and capabilities of other users are obscured (Kloess & van der Bruggen, 2021). Because there is no centralized, legitimate authority to enforce contracts or verify the quality of the information being traded, interactions are inherently fraught with risk.

Information Asymmetry and "The Market for Lemons"

The lack of verifiable intelligence creates an environment heavily skewed by information asymmetry. In economics, when buyers cannot reliably verify the quality of a product, the ecosystem devolves into a "market for lemons," where low-quality goods overrun the market (Sebagh et al., 2022). In underground forums, vendors are incentivized to exaggerate the effectiveness of their data (e.g., stolen credentials, vulnerabilities) to maximize profit. Scammers and low-skill participants continuously attempt to sell obsolete or patched exploits, relying on the fact that buyers cannot test the product without executing the crime (Campobasso et al., 2023).

The "Silent Failure" Hypothesis

This paper introduces the concept of the "silent failure" to explain how redundant criminal attempts occur. If Offender A accesses a shared data pool, identifies a target, and initiates a crime script that ultimately fails, rational choice theory dictates that they must absorb the cost $c(F)$.

However, Offender A possesses virtually no incentive to report this failure back to the data pool, driven by several structural disincentives:

1. **Reputational Damage:** Admitting failure in peer-moderated networks can degrade an offender's status, signaling incompetence rather than flawed data (Kloess & van der Bruggen, 2021).
2. **Economic Disincentive:** If the offender paid for the script, admitting it was useless provides no financial recourse, as illicit markets lack consumer protection.
3. **Loss of Leverage:** If the offender intends to repackage and resell the data to recoup their

losses, revealing that the target is secure would destroy the data's resale value (Campobasso et al., 2023).

Because the failure is essentially silenced, the shared data pool remains unmodified. Offender B, accessing the identical data pool, reviews the same intelligence. Because $p(F)$ remains artificially minimized due to the missing failure feedback, Offender B calculates a positive expected utility and executes the exact same failed script. This cycle persists until the target's defenses are explicitly broadcasted by external entities (e.g., law enforcement press releases) or the data ages out of relevance.

Implications for Disruption Strategies

Understanding the mechanics of silent failures offers actionable avenues for situational crime prevention. Traditional interventions rely on infrastructure takedowns and arrests, which are resource-intensive and frequently result in temporal or spatial displacement of the market (Sebagh et al., 2022).

Alternatively, defenders can actively exploit the structural vulnerability of information asymmetry. By deliberately injecting "failure feedback" or artificial uncertainty into criminal data pools, authorities can fundamentally alter the rational choice calculations of prospective offenders. Tactics aimed at attacking user reputation or highlighting the unreliability of shared scripts raise the perceived $p(F)$, successfully eroding the trust required to facilitate collaborative crime (Sebagh et al., 2022).

Conclusion

Criminal networks are highly reliant on shared data pools to coordinate operations, yet they are structurally incapable of efficiently managing operational failure. The lack of a failure feedback loop ensures that unsuccessful attempts remain invisible to the broader network. Consequently, successive actors are mathematically incentivized to repeat the same failed actions against the same targets. Recognizing this dynamic not only provides theoretical insight into offender decision-making but also highlights strategic opportunities to disrupt illicit networks by weaponizing their inherent information asymmetry.

References

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